INDEPENDENT CHECK · NOT PEER REVIEWED

Do gravitational waves decay into light?

An independent check of D. Blas and J. A. Oller, “Gravitational waves decay in vacuum,” arXiv:2607.24930 (version 2)

Prepared for Ethan Van Winkle · 27 October 2026 · From the Science repository, github.com/EthanDJohnson/Science, folder `checks/2026-10-11-blas-oller-gw-decay/`

HOW THIS WAS MADE

The check was run by AI agents (Anthropic’s Claude models), directed from the Science repository. Several independent analyses attacked the paper from different directions, adversarial referees challenged each one, and a final critic re-ran the scripts. Every number below was printed by a script whose output log is kept beside it. No physicist has reviewed it yet. Read it as a careful draft of a referee report, not as a published result. Where it is uncertain, it says so.

The short version

- **The claim.** Treat a gravitational wave as a coherent state of N gravitons. Two gravitons can fuse into two photons (g g → γγ). Each fusion is suppressed by G^2, but the rate grows as N^2, so the authors find that gravitational waves decay into photon pairs in empty space, with consequences for binary mergers, the cosmic microwave background (CMB) and dark matter.
- **Their algebra checks out inside their model.** We reproduced their amplitudes by three independent routes, their key coefficient $3/(25\pi)$ exactly, their stochastic-background rate $8\pi G^2\rho^2/15$, and their $\hbar$ bookkeeping.
- **But their model is not a wave travelling away from its source,** for two separate reasons:
  1. The amplitudes are complex numbers whose phases they drop. With the phases kept, the rate is 0.25–0.60 of theirs for realistic polarisation patterns, and exactly zero for a circularly polarised wave.
  2. The quantum state they use carries an *incoming* spherical wave that collapses onto the source and then re-expands. About 85% of their probability is generated at that collapse point, within one wavelength of the source.
- **A real outgoing wave makes photon pairs only near its source.** The number made beyond a radius r falls as $1/r^3$. So there is no decay in flight, no “reaction time,” and no depletion of gravitational-wave backgrounds over cosmic distances.
- **Not settled:** production right at the source, inside about $\lambda/2\pi$, where a crude estimate is 0.3–0.7 of their number for a black-hole merger. So we do *not* claim their merger numbers are orders of magnitude too large. We claim they describe production at the source while it radiates, not a gravitational wave decaying in vacuum.

1. What Blas and Oller claim

Blas and Oller ("B&O" below) model a gravitational-wave front as a coherent state of N gravitons of frequency ω and bandwidth σ. Their abstract opens: "We show that gravitational waves (GW), treated as coherent graviton states, decay into photon pairs in vacuum. The process, even if suppressed by G^2, is lifted by two effects combined: the expected factor of the graviton number squared, N^2, and the coherence of the state."

The ingredients are the tree-level amplitudes for two gravitons going to two photons (their eq. 6), with magnitudes $8\pi G\, t^2/s$ and $8\pi G\, u^2/s$ in terms of the Mandelstam variables, summed coherently over the wave's graviton modes (eq. 7). The result is a pair-production rate (eq. 8), in natural units ($\hbar = c = 1$):

$$dP/dt = 3\, G^2\, N^2\, \omega^3\, \sigma^2 / (25\pi) \qquad \text{B\&O eq. 8}$$

From it they derive a reaction time τ, the time to the first photon, and a depletion time for the wave (eqs. 9–12); a photon luminosity for compact binaries (their Fig. 2); a rate per volume for a stochastic background, $8\pi G^2 \rho_{GW}^2/15$ (eq. 14); an enhancement by CMB photons already present, a factor $1 + 2f$ with f the CMB occupation number; a cosmological integration over the history of all mergers (eq. 19 and supplement eq. S12); and the analogous decay into ultralight dark-matter scalars (eqs. 16–18).

2. How we checked it

Four analyses, each with its own scripts:

1. **Amplitudes and helicity phases.** We rebuilt the full complex amplitude three ways that share no code: a new amplitude module, a spinor-helicity "double copy" (KLT) closed form, and a covariant route with no spinors. The first two agree to 10^{-12} for random kinematics; the third agrees to 10^{-14} head-on and 9×10^{-9} over 200 random configurations.
2. **The space-time structure of the state.** What field the coherent state actually carries, and where in space and time the pair probability is generated.
3. **The $2\pi\delta(0) \to T$ step**, which turns a probability into a rate.
4. **The stochastic background, CMB stimulation and ultralight dark matter.**

Adversarial referees then tried to break each analysis, and a critic re-ran the scripts and checked the report against the logs. We quoted B&O only from their own text, extracted from the PDF.

3. What holds up

Inside B&O's idealised model, the following are right:

- the **magnitudes** of the eq. 6 amplitudes, and the pattern of which ones vanish;
- the **eq. 8 coefficient** 3/(25π), exactly, from their own integral;
- the **stochastic-background rate** (eq. 14), $8\pi G^2\rho^2/15$ per unit volume, for any isotropic unpolarised spectrum, and their conclusion that it is utterly negligible (10^{-110} to 5×10^{-109} photons per cubic metre per second, depending on the background's size);

- the **drop-out of coherent intermediate gravitons**, and the scalar amplitudes and coefficients (7/(450π), 1/16) for the dark-matter channel;
- the **ħ counting**: the pair number contains no ħ at fixed classical wave energy and frequency. It is a quartic functional of the wave's classical strain, so the celebrated N^2 is ordinary classical phase coherence, not a quantum bonus.

The problems are not arithmetic. They are about whether the quantity computed is the one that describes a real gravitational wave.

4. Problem 1: the helicity phases are dropped

Scattering amplitudes are complex numbers, and for g g → γγ their phases depend on the directions of the outgoing photons. B&O's eq. 6 lists magnitudes whose labels are spin projections on a common axis rather than physical helicities. That is fine for the magnitudes, but B&O's eq. 7 then adds the amplitudes as *real positive numbers*. A combination of them that doesn't depend on phase conventions has phases of +63.9°, −17.1° and −125.8°, so no choice of convention makes them all real at once.

Keeping the phases matters. Two angular-momentum channels (J_z = +4 and −4) cannot interfere once you integrate over the photons' azimuth, because their cross term goes as cos 8φ and averages to zero. B&O's rule keeps that cross term as a constant. As a result:

- for a pair of head-on gravitons the rate is 6.4 (in the check's units), not B&O's 7.47;
- the rate depends wrongly on the relative orientation of two linear polarisations (5.33 instead of the physical 6.4 for plus × cross);
- read literally, eq. 7 gives a nonzero rate for two right-circular gravitons, where the physical rate is exactly zero.

B&O's "unpolarised isotropic" wave has no unique physical realisation (a smooth polarisation pattern cannot be combed flat over a sphere, the hairy-ball theorem). So we evaluated several real patterns:

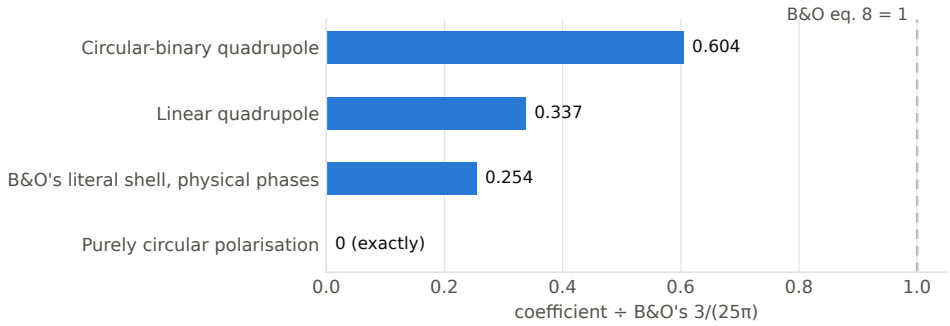


Figure 1. The eq. 8 coefficient for physical coherent shells, as a fraction of B&O's 3/(25π). An independent integrator agrees to 0.01% (to 0.2% for the literal shell, which converges slowly). For the same linear-quadrupole pattern, B&O's own rule gives 0.891 instead of 0.337. Sources: `notes/taskA3b_smooth.log`, `notes/taskA3b_bo_plus_grid2.log`.

Eq. 8 therefore overestimates by 1.7× to 4×, depending on the pattern. A smaller, separate inconsistency: eq. 5 gives the wave a Gaussian profile, but eqs. 7 and S6 are evaluated with a top hat. At equal N and σ the Gaussian gives a 4π times larger peak rate and a 3.76 times larger total, so the 3/(25π) belongs to the top hat specifically.

5. Problem 2: the wave in the calculation is coming in, not going out

B&O compute an S-matrix probability with the initial state $|f\rangle = S_{source}|0\rangle$: the coherent state that the source would create, used as the incoming state of free fields. A free field that agrees with the source's outgoing radiation at late times must, at early times, have been an *incoming* spherical wave. That is the Pauli–Jordan (retarded minus advanced) solution. In the calculation the wave converges onto the source's position, focuses there at $t = 0$, and re-expands. Numerically, the field is 100% incoming at early times ($t\omega = -30$) and 50% incoming at the focus. A real source's retarded field is never incoming.

That focus is where B&O's probability comes from:

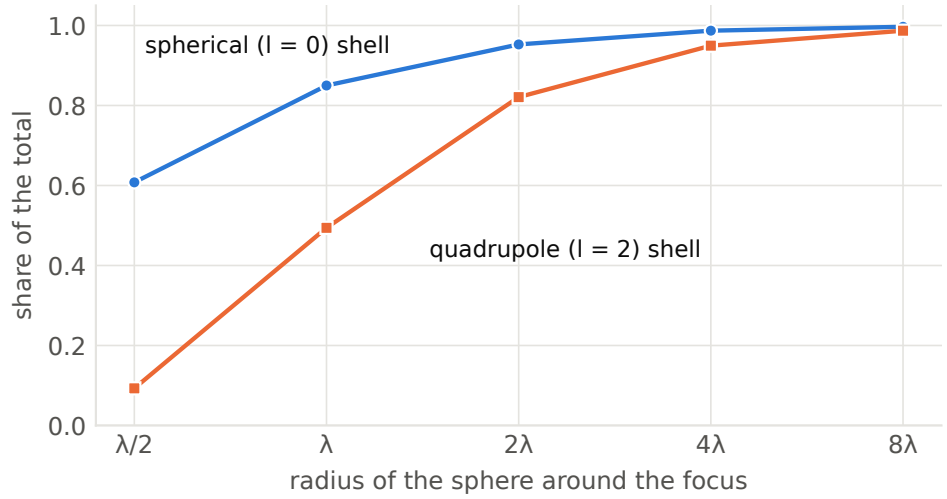


Figure 2. Share of B&O's total pair probability generated inside a sphere of the given radius around the focus. For a spherical shell, 85% comes from within one wavelength; removing everything inside one wavelength leaves 2%. A quadrupole shell is more spread out but still local (82% within two wavelengths). B&O's "effective volume" $5\lambda^3/(9\pi\varepsilon^2)$, 17.7 λ^3 at their $\varepsilon = 0.1$, is this focal region. Source: `notes/taskB_where_grid.log`.

Why the focus matters: collinear gravitons can't fuse. Two massless particles moving in the same direction have zero invariant mass, $s = 0$, and a pair of photons needs $s > 0$. Far from its source a wave is locally almost a plane wave, all its gravitons nearly parallel, so $s \approx 4\omega/r$ is tiny and fusion shuts off. At the focus, gravitons arrive from every direction, s is large, and fusion is easy. It is the same kinematics that stops a single photon from turning into an electron–positron pair in empty space.

To be fair to B&O: they note that far from the source the field tends to the outgoing packet, which is true at late times. They also use the Feynman propagator correctly for the internal graviton. The problem is the external state: the S-matrix integrates over all times, including the fictitious incoming half of the wave.

6. Problem 3: a total, not a rate

B&O turn the energy-conserving delta function into a time by the textbook replacement $2\pi\delta(0) \to T$. Kept exactly instead, it gives a finite total for the whole packet,

$$P_{total} = (4/25)\, G^2\, N^2\, \omega^3\, \sigma \text{ pairs},$$

produced with a time profile $R(t) = R(0)\,\mathrm{sinc}^4(\sigma t/2)$, where $R(0)$ is exactly B&O's eq. 8. So eq. 8 is the *peak* rate, sustained for an effective time $4\pi/(3\sigma)$.

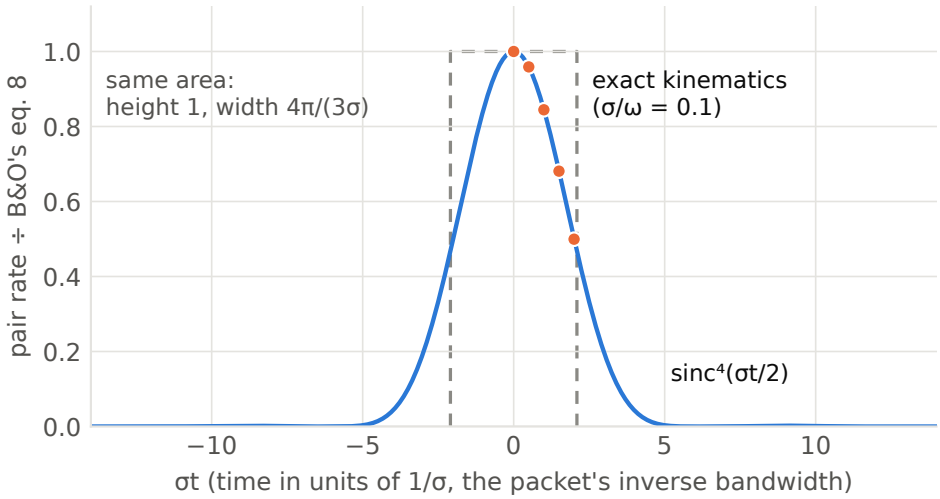


Figure 3. Pair production in B&O's own state happens once, in a burst lasting about one inverse bandwidth. The curve is $\mathrm{sinc}^4(\sigma t/2)$, the dots an exact-kinematics calculation ($\sigma/\omega = 0.1$), and the dashed rectangle has the same area. Source: `notes/taskC1.log`.

As an average over the packet, B&O's idealisation is legitimate. What breaks is using it as a constant rate over times much longer than $1/\sigma$. Eq. 9 exceeds the total for $t > 4\pi/(3\sigma)$, and in B&O's own model the fraction of a packet that is ever converted is $(4\pi/3)/(\sigma\tau_h)$. For their own example ($E = M_\odot c^2$, $\varepsilon = 0.15$, $\omega = 10^4$ eV) that fraction is 2.4×10^{-36}. That is not depletion.

7. What a real outgoing wave does

To model a physical wave, we kept only the outgoing half of B&O's packet, switched on beyond a radius $r_{\min}$, and computed the pairs it produces with the full in-in (expectation-value) formalism:

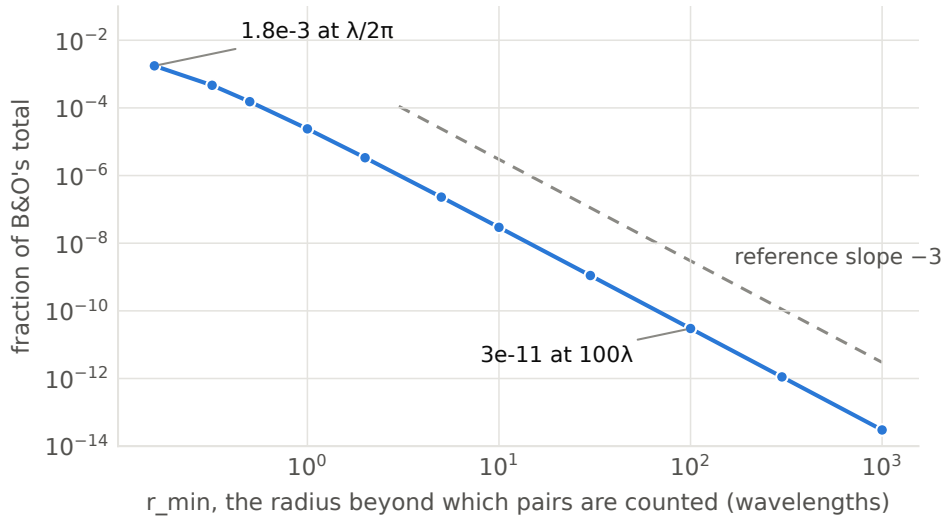


Figure 4. Pairs produced beyond r_{min} by the outgoing wave, as a fraction of B&O's total. It includes a coherence factor of 0.41 that only one module computes, which shifts the curve but not its slope. The local slope tends to −3.000; three different switch-on shapes change the prefactor by up to 2.2× but not the slope. Source: `notes/taskB_outgoing.log`, section (h).

The rate while the shell passes radius r falls as r^{-4}, and a quadrupole wave with its near-field terms behaves the same way. This is a formation-length effect, and it undoes the claims that rely on a sustained decay:

- **No reaction time and no depletion** (eqs. 9–12; their footnote 4, which pictures photons produced on a "sphere of radius τ").
- **The cosmological integration (eq. 19/S12) overcounts.** It lets each merger radiate photons for the whole propagation time. In B&O's own model the energy per event is the rate times 4π/(3σ), so S12 overcounts by 3σ/(4πH), for example 1.7×10^{21} at $\omega = 10^{-10}$ eV today.
- **The "lower bound" from binning is an overestimate.** Splitting a chirping signal into frequency bins and using eq. 8 per bin overstates the exact result inside B&O's allowed region. The overstatement grows with the number of wave cycles in a bin. How large it is inside B&O's own validity region is contested: around order one at the typical point, up to about 7,000 at the extreme.

Worked example: a GW150914-like merger

$E = 2\ M_\odot c^2$ radiated at 150 Hz over 5 ms (SI units):

Quantity	Value
B&O eq. 8 rate, applied to the merger	1.40×10^{-4} pairs/s
B&O reaction time τ (eq. 10)	3.58×10^{3} s (720,000 times the merger's duration)
B&O rate × 5 ms	7.0×10^{-7} pairs
Outgoing wave, pairs made beyond λ/2π (318 km)	1.9×10^{-9} pairs (1.4 with the CMB enhancement)
Outgoing wave, beyond one wavelength (2,000 km)	2.6×10^{-11} pairs (0.02 with the CMB enhancement)
Outgoing wave, beyond ten wavelengths	3.2×10^{-14} pairs

Near zone: quadrupole wave with its $1/r^2$ and $1/r^3$ terms, beyond $\lambda/2\pi$	$2.1–4.9\times10^{-7}$ pairs, i.e. 0.30–0.70 of B&O

The last row is why we don't claim B&O's merger numbers are hugely too large: close to the source the numbers are comparable. But $\lambda/2\pi = 318$ km is about the binary's own separation (339 km), inside the source, which none of these models describe. B&O's validity condition $\tau \ll 1/\sigma$ also fails in vacuum and holds only with the CMB enhancement, where $\tau = 4.7$ µs and their "sphere of radius τ" has a radius of 1.4 km, again deep inside the source.

8. Verdict on each claim

"Right arithmetic, wrong quantity" means the calculation is correct but does not describe a gravitational wave travelling away from a real source.

B&O claim	Verdict	Key number
Eq. 6 amplitude magnitudes and zero pattern	**Confirmed**	as spin projections on a common axis; agree to 10^{-10}
Eq. 6 amplitudes usable as real positive numbers (as eq. 7 needs)	**Refuted**	invariant phases +63.9°, −17.1°, −125.8°
Eq. 8 coefficient $3/(25\pi)$, within the model	**Confirmed**	exact (sympy); numerical ratio 1.000000000000
Eq. 8 for physical coherent waves	**Right arithmetic, wrong quantity**	0.25–0.60 × eq. 8; 0 if circular
Coefficient set by N, ω, σ alone	**Refuted**	depends on profile (4π) and polarisation
$2\pi\delta(0) \to T$	**Confirmed as an average**	eq. 8 is the peak; $T_{eff} = 4\pi/(3\sigma)$
Field "tends asymptotically to the outgoing" packet	**Right arithmetic, wrong quantity**	true late; the in-state is incoming early
Where the eq. 8 probability is generated	**At the focus**	85% within 1λ (spherical shell)
Sustained decay, depletion, "sphere of radius τ"	**Refuted**	yield beyond $r \propto r^{-3}$
Photon luminosity of binaries (Fig. 2)	**Right arithmetic, wrong quantity**	near-source comparison undetermined
Binning gives a lower bound	**Refuted**	it overestimates; size contested
Eq. 19/S12: each merger radiates for the whole propagation time	**Refuted**	overcount $3\sigma/(4\pi H)$, e.g. 1.7×10^{21}
$\hbar$ counting (eqs. 11–12) and the N^2 enhancement	**Confirmed**	N^2 is classical phase coherence
Eq. 14, stochastic background $8\pi G^2\rho^2/15$	**Confirmed**	exact for any isotropic spectrum
CMB factor 1 + 2f "in the average sense"	**Confirmed as an approximation**	exact average 5–8% higher
Dark-matter scalar coefficients $7/(450\pi)$ and $1/16$, within the model	**Confirmed**	physical states 0.20–0.47 of it
Eq. 18, scalar power ~10^{14} W	**Arithmetic right, applicability doubtful**	3.8×10^{13} W; resonant band ~10^{-6} of the merger's

The full 29-row table, with the script behind every number, is section 3 of `report.md` in the check folder.

9. What is not settled

1. **Production right at the source** (within about λ/2π) is not computed. It needs the binary's own near-zone field and the coupling of matter to gravitons. This is why we call the comparison with B&O's binary luminosity undetermined.
2. **The retarded-field argument is analytic only.** That the physical field is the retarded, outgoing one when the source is kept inside the calculation was argued, not checked numerically.
3. **Counting pairs "beyond r_{min}" depends on how the wave is switched on,** by up to a factor of a few. The r^{-3} fall-off does not.
4. **There is no unique physical polarisation pattern** for B&O's isotropic idealisation. The patterns we tried give 0.25–0.60, and a circularly polarised wave gives 0.
5. **The finite-mass scalar channel is incomplete:** its physical interference term was not computed.

Where this could go next

You'll know better than we do. Some options:

- **A sanity check on Problem 2 by someone who works with QFT in curved or radiating backgrounds.** It is the crux: if the in-state argument is wrong, most of the rest falls back to B&O's favour.
- **Writing to the authors** with the specific points (Sections 4–7), or a short "Comment on" paper. A comment on arXiv needs an endorser in the field.
- **Finishing the open piece:** production inside the source region, with a real binary model.
- **In the classroom,** the collinear-gravitons argument in Section 5 may be the most usable piece. It is the same reason a lone photon can't become an electron–positron pair in empty space, and it explains why a wave far from its source is stable.

Reproducing the numbers

Everything is in `checks/2026-10-11-blas-oller-gw-decay/`: `report.md` (the full report, quoting B&O verbatim), `calc/` (about 35 Python scripts) and `notes/` (their output logs). The scripts need numpy, scipy, sympy and mpmath. Some import the repository's `qft_second_order.py` and `qft_production.py`, and every module output the conclusions rest on is cross-checked by an independent route. Run them from the check folder, for example `python3 calc/taskC1_exact_Ptot.py`. A few take several minutes. The figures here are drawn by `summary/make_figures.py` from the logged numbers.

Glossary

Coherent state
The quantum state that best resembles a classical wave: a superposition of graviton numbers with a definite average field. A classical gravitational wave is a coherent state with an enormous N (about 3.6×10^{78} for the merger example).

Helicity

A massless particle’s spin along its direction of motion: ±2 for gravitons, ±1 for photons. Circular polarisation means one helicity.

In-out versus in-in

An S-matrix (“in-out”) calculation asks for the probability to go from a state in the far past to one in the far future. An “in-in” calculation asks for the expectation value of an observable, here the number of photon pairs, given the past state. For radiation from a real source, the in-in route with the retarded field is the physical one.

Pauli–Jordan function

The retarded minus the advanced solution. It is a source-free solution of the wave equation, so it can describe a free wave, and it contains an incoming half as well as an outgoing one.

Formation length

The distance a process needs to “complete.” For nearly parallel gravitons the invariant mass is small, the formation length is long, and pair production far from the source is suppressed.

Natural units

$\hbar = c = 1$, so G has units of inverse energy squared and every quantity can be expressed in powers of energy.